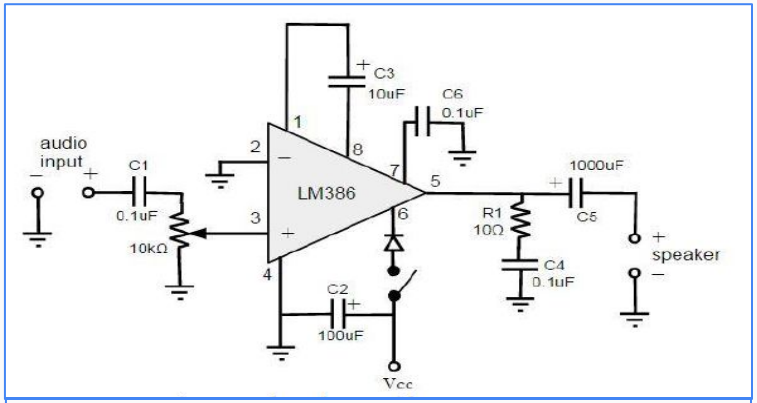
Lab 2: Analog Circuits

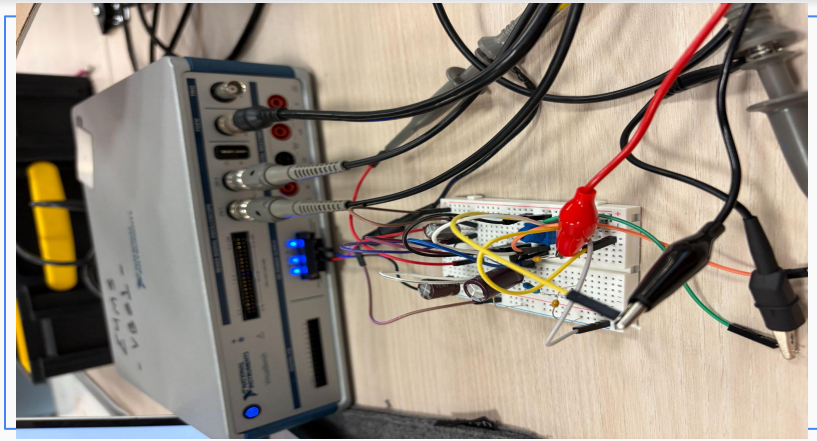
Tanzim Rahman
Nizar Ali
ECE 5 - UC San Diego

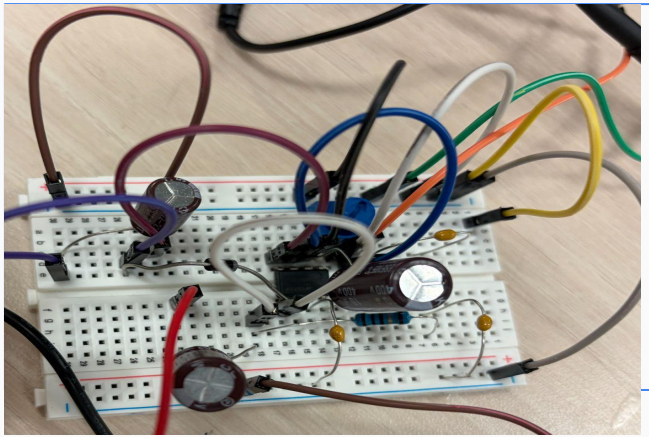
Part 2: Audio Amplifier

Challenge #3a - Audio Amplifier (Breadboard Circuit)

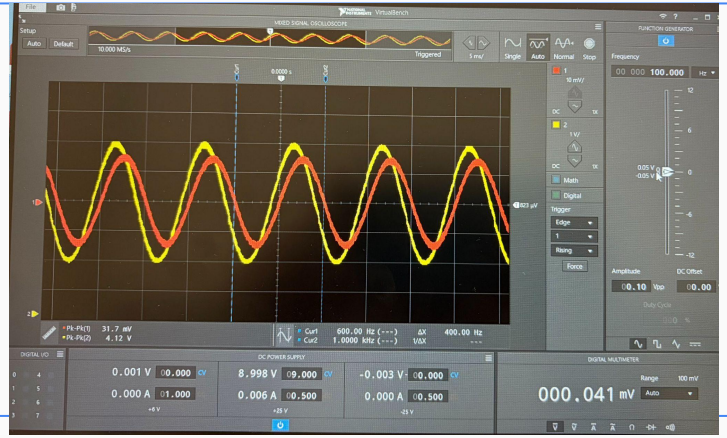


Schematic of Audio Amplifier Circuit

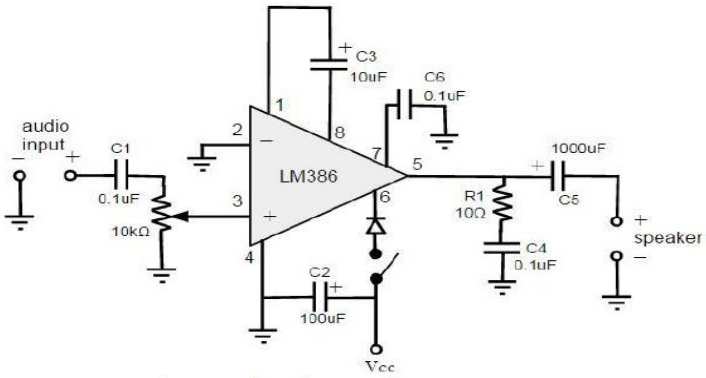




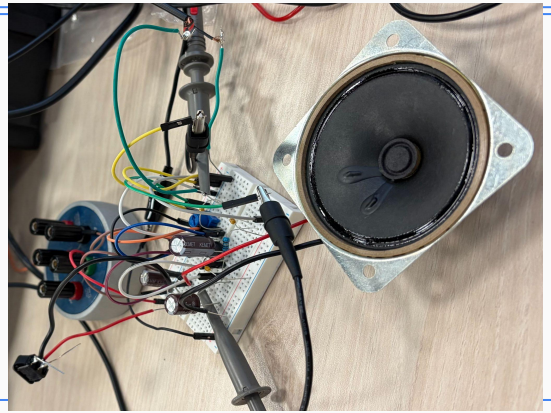
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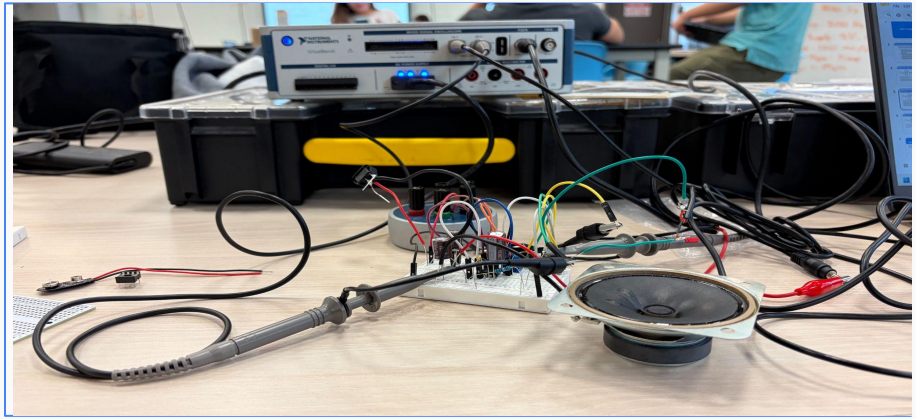


Challenge #3b - Audio Amplifier (Breadboard with Soldered Parts)



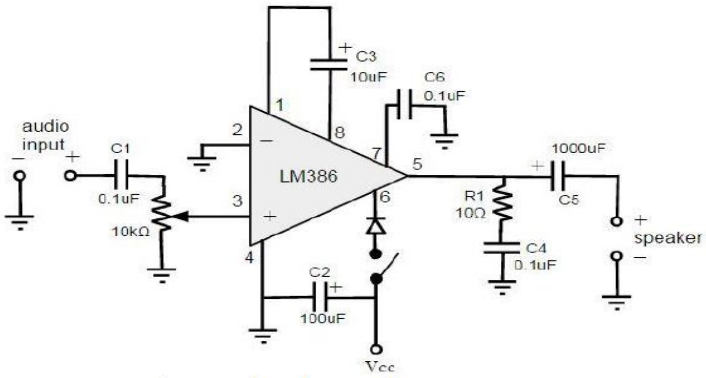
Schematic of Audio Amplifier Circuit





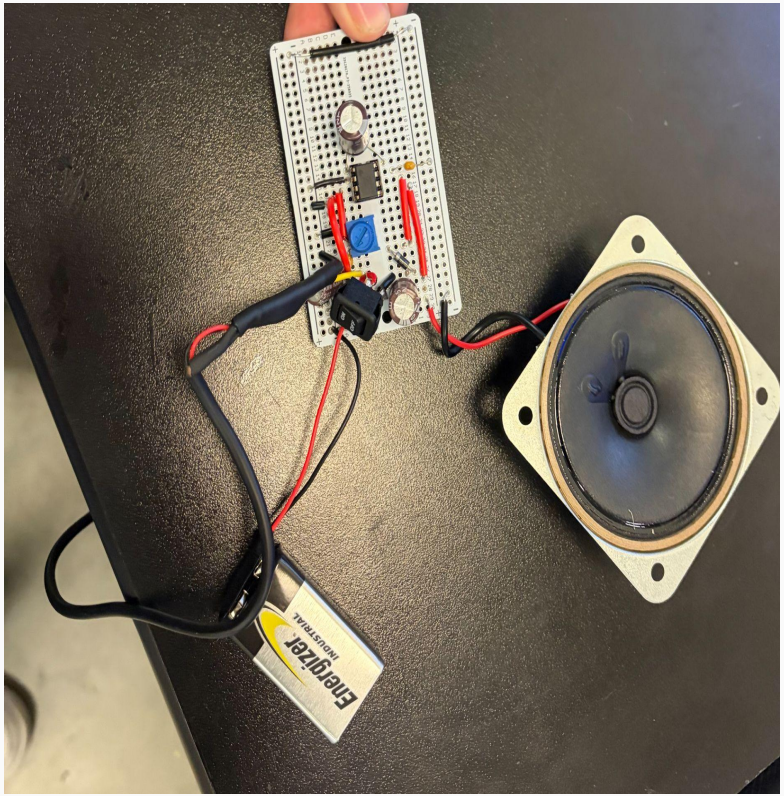


Challenge #3c - Audio Amplifier (Fully Soldered)



Schematic of Audio Amplifier Circuit

This is a audio amplifier circuit which amplifies any frequencies passed to it. The amplitude of the frequencies are controlled by the potentiometer. On the first go, we shorted our soldered circuit On the second go, we redid it and became very diligent with the wiring. This caused the second attempt to be successful. We tested it on a breadboard to ensure the circuitry was right before we soldered anything.



Lab 2 Equipment and Experience

Explain the purposes for each of the following lab equipment: multimeter, power supply, oscilloscope, function generator?
We used the multimeter to measure continuity and ensure the proper quantity was following through a given element. The power supply supplied our circuit with the voltage necessary to have it run. The oscilloscope measure the amplitude and the frequency of different elements in our circuit, and the function generator is what created the waves.

The most valuable thing to me was shorting the soldered circuit. It was extremely frustrating, but understanding why it got shorted helped the second iteration be much cleaner. I think getting a hands-on experience of how a function generator and amplifier play out in our day-to-day lives (speakers) really helped me conceptualize this lab. Even though the soldering took us around eight hours total, I totally enjoyed the feeling at the end of accomplishing something a little more complex than usual.